E-Commerce Sales Performance and Customer Insights Dashboard

1. Project Overview and Objective

To analyze e-commerce sales data and develop interactive Power BI dashboards that provide actionable insights into sales performance, customer behavior, product and category performance, and regional sales trends. The project aims to transform raw data into meaningful visualizations, enabling stakeholders to monitor key performance indicators (KPIs), identify trends and opportunities, and support data-driven business decision-making.

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2. Data Sources

- **Source Description and Timeline:** Google Dataset Search **and** 2020-2024.
- **Domain**: E-commerce / Sales Analytics

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3. Problem Statement

- To analyze overall sales performance across products, categories, and regions to identify revenue trends and business growth opportunities.
- To study customer purchasing behavior and buying patterns for improved customer engagement and retention.
- To evaluate product and category performance to identify best-selling and underperforming products.
- To monitor key business metrics through interactive dashboards for effective performance tracking.
- To identify regional sales trends and market opportunities to support strategic business decisions.
- To transform raw e-commerce sales data into meaningful visualizations that enable data-driven decision-making.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Customer ID	Integer / String	Unique identifier for each Customer
Customer Name	String (Text)	Name of the customer
City	String (Text)	City where the customer resides
State	String (Text)	State where the customer resides
Country	String (Text)	Country where the customer resides
Order ID	String (Text)	Unique identifier for each order
Order Date	Date	Date when the order was placed
Product ID	Integer / String	Unique identifier for each product
Product Name	String (Text)	Name of the Product
Category	Categorical	Category of the Product
Brand	String (Text)	Brand of the Product
Quantity	Numeric (Integer)	Number of units ordered
Unit Price	Numeric (Decimal)	Price per unit of the product
Discount	Numeric (Decimal)	Discount applied to the order
Tax	Numeric (Decimal)	Tax charged on the product
Shipping Cost	Numeric (Decimal)	Shipping cost of the order
Total Amount	Numeric (Decimal)	Total amount paid for the order
Payment Method	Categorical	Method used for Payment
Order Status	Categorical	Current status of the order
Seller ID	Integer / String	Unique identifier for the seller

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Standardized column names:** Column names were renamed to improve readability and ensure a consistent naming convention (e.g., Customer ID, Customer Name).
- **Standardized column formats:** Column formats were updated based on the nature of the data to ensure proper representation and consistency. Date values were formatted using the Date format, monetary fields such as Unit Price, Tax, Shipping Cost, and Total Amount were formatted as Currency, and percentage-based fields such as Discount were formatted as Percentage.
- **Removed duplicate records:** A duplicate record was identified in the Order Details sheet using Conditional Formatting and removed using the Data → Remove Duplicates feature. .
- **Handled missing Product IDs:** Missing values in the Product ID column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct Product ID based on existing records.
  =IF(ISBLANK([@[Product ID]]),XLOOKUP([@[Product Name]],FILTER(D:D,C:C<>" "),FILTER(C:C,C:C<>" ")),[@[Product ID]])
- **Handled missing Product Names:** Missing values in the Product Name column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct product name based on existing records.
  =IF(ISBLANK([@[Product Name]]),XLOOKUP([@[Product ID]],FILTER(C:C,D:D<>" "),FILTER(D:D,D:D<>" ")),[@[Product Name]])
- **Corrected Product Category values:** The Product Category column contained incorrect values. To resolve this, the unique Product Name and Product Category values were extracted using the UNIQUE() function and organized into a separate worksheet named Product Category. The XLOOKUP() function was then used to retrieve and update the correct category for each product.
- **Standardized Customer Names:** The Customer Name column in the Customer Details table contained inconsistent letter casing and extra spaces. These issues were corrected using a combination of the PROPER() and TRIM() functions to standardize the formatting.

- **Handled missing State values:** Missing values in the State column were identified and replaced using a combination of the IF, XLOOKUP, and FILTER functions to retrieve the correct state based on existing records.
  =IF([@State]="",XLOOKUP([@City],FILTER(C:C,D:D<>""),FILTER(D:D,D:D<>"")),[@State])
- The Country column contained inconsistent values (e.g., *United States*, *United States*, *US*, and *us*). These inconsistencies were identified using the Filter feature and corrected using the Find and Replace tool to ensure a consistent country name throughout the dataset.
